

ETHIRAJ SRINIVASAN

CONTACT

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JOB OBJECTIVE

Aspire to lead data-driven innovation by architecting scalable platforms, fostering engineering excellence, and aligning technology strategy with organizational growth and long-term vision.

CORE COMPETENCIES

- Strategic Data Platform Planning
- Big Data Analytics & Processing
- Data Pipeline & ETL Architecture
- Real-Time & Batch Processing
- Data Quality & Governance
- Cloud Data Solutions
- Performance Optimization & Resource Management
- Analytics & Reporting
- Team Leadership & Mentoring
- Cross-Functional Collaboration
- Ingestion Application Development
- Technical Innovation & Strategy

AWARDS

- Shopee, Singapore - Value Star Award
- DNA Capitals, Singapore - Winner - Hackathon
- Tata Consultancy Services - Star of the Learners group, On the Spot Award

EDUCATION

- 2018: National University of Singapore, Singapore - M.Tech Software Engineering, CGPA - 4.36/5
- 2013: Anna University, India - B.E Electronics and Communication, CGPA - 8.56/10

PROFILE SUMMARY

- **Strategic Data Engineering Leader** with a **decade of experience** driving **enterprise-scale data platforms, architecture, and analytics modernization** across diverse industries. Possesses deep expertise in **data pipeline design, distributed processing, ETL, and real-time and batch frameworks**, with a proven ability to deliver **scalable, high-performance, and reliable data ecosystems**.
- Leads and mentors **cross-functional teams** in building and managing **data ingestion, ETL, and analytics platforms**, ensuring strong **data governance, data quality, observability, and operational excellence**. Skilled in designing and maintaining **data lakes, data warehouses, ML-ready datasets, and real-time analytics pipelines** to support enterprise **decision-making**.
- Expert in implementing **data quality frameworks, monitoring systems, and cost-optimized solutions** to improve **reliability, reduce latency, and enhance efficiency**. Proficient in **Spark, Hadoop, Kafka, Hudi, Hive, HBase, AWS (EMR, S3, Athena), Python, Spring Boot, Redis**, and other modern **data technologies**. Recognized for combining **technical depth** with **strategic vision** to drive **data-driven innovation and organizational growth**.

WORK EXPERIENCE

Aug'25 – Present | Zgrow Solutions, Chennai | Head of Data Engineering

Stack - Spark,S3,Athena,EMR

Key Result Areas:

- **Led Data Engineering**, mentoring a **high-performing team** and implementing **governance, best practices, and process optimizations** to deliver **scalable, reliable, and high-quality data platforms**.
- **Architected an end-to-end Spark-based ML pipeline on AWS EMR for logistics anomaly and weight prediction**, integrating a **distributed ML model** and **automated feature engineering**, achieving **99.9% uptime**.

Jan'25 – May'25 | Synechron, Chennai | Lead Data Engineer

Stack - Spark, Hadoop, Kafka

Key Result Areas:

- **Led the development** of a **high-performance reconciliation platform** by building a **Kafka-to-HDFS data connector** and actively working on core components such as the **data merger, preprocessor, reconciliation engine, and audit logger**, ensuring accurate and traceable validation across large-scale datasets.

Jun'20 – Feb'25 | Shopee, Singapore | Expert Data Engineer

Stack - Spark, Hadoop, Hudi, Hive, Kafka, HBase, Python, Redis, Docker, Spring Boot, Ruby on Rails

Key Result Areas:

Batch Ingestion Team Leadership:

- Led the batch ingestion team managing **23k MySQL-to-Hive pipelines, 6k Hive-to-Hive pipelines, and 2k Hive-to-HBase and HBase-to-Hive pipelines** with daily and hourly frequencies.
- Ensured seamless operations on a 14PB data lake, achieving an outstanding **99.95% job success rate for 31k ingestion pipelines**.

Performance Optimization & Resource Utilization:

- Maintained **80% memory utilization**, processing **1,059,221 GB/hour** and **245,414 Vcore/hour**, with **97% of jobs completing within 30 minutes**.
- Optimized cross-IDC Hive-to-Hive pipelines, saving approximately **94,325 GB/hour, 13,487 Vcore/hour, and 500 TB of storage**.

TALKS & ENGAGEMENTS

- Jury - HackElite' 25 - SRM Institute of Science and Technology
- Guest of honor - International Conference on Computer Communication & Power Systems - ICCCPs 2020
- Guest of honor - Outcome Based Industry 4.0 at Sri Venkateswara College of Technology
- Resource Person - Faculty Development Program On Artificial Intelligence For Robotics (AICTE Training and Learning (ATAL Academy) At Sai Ram Engineering College

OPEN SOURCE CONTRIBUTIONS

- Ruby Gems
 - Collaborator - Sunspot (Apache Solr) gem
 - Owner - Collection2csv, devise-foundation-views, rails-fort, rails-protip

PERSONAL DETAILS

- DOB: 3rd Oct 1991
- Languages: English, Telugu & Tamil
- Address: Chennai

Platform Upgrade & Cost Optimization:

- Upgraded the ingestion platform to **Spark 3**, resulting in a **9% reduction in processing time** and a **14% reduction in resource usage**.

Pipeline Development & Monitoring:

- **Designed and monitored batch, real-time, and cross-country ingestion pipelines**, including **hourly Hive-to-Hive jobs** with **filters, incremental dumps, and Data Quality Checks (DQC)**.

Data Pipeline & Metrics Management:

- Designed metrics for job performance and resource utilization, and managed ingestion pipelines handling diverse sources (**MySQL, CSV, Kafka**) and sinks (**HDFS, Hive, Hudi, Kafka**) using **Spark, DeltaStreamer, Spring Boot, and Maxwell**.

Ingestion Applications Development:

- Built and maintained key applications: **Core Engine** (job automation), **AutoAdaptation Framework** (dynamic optimization), **Operations Platform, Data Quality Module, Configuration Service, and Notification Service**.

Feb'19 – May'20 | Lomotif, Singapore | Data Engineer

Stack - Spark, Kinesis, Lambda, EMR, Athena, Quicksight, S3, Python, Redis, Elasticsearch, Postgres, Flask

Key Result Areas:

- **Data Infrastructure Development:** Led the design of a scalable **data infrastructure platform**, including a **data lake** using **S3, Kinesis, Lambda, and EMR (Spark)**, and implemented an **automated recommendation engine** to enhance feeds and follower engagement.
- **Big Data Processing & Analytics:** Built and monitored **PySpark jobs on EMR** for diverse business use cases, developed advanced **metrics for ranking videos**, and conducted **graph analysis** to identify communities and influencers.
- **Real-Time Visualization:** Created interactive **QuickSight dashboards** with **Athena** as the query engine, providing actionable insights for stakeholders and enabling data-driven decisions.

May'18 – Jan'19 | Knorex, Singapore - Big Data and Analytics Engineer

Stack - Spark, Kinesis, Lambda, S3, Scala, Vertx

Key Result Areas:

- Migrated an ad events tracker to **Vert.x 3**, achieving **50K QPS** with optimized throughput using **Kinesis and Lambda architecture**; built a **data lake** on **S3** integrated with real-time sources, developed **PySpark jobs** on **EMR** for analytics.
- Created a **reporting tool** with **Spark (Scala)** for attributions, and designed **QuickSight dashboards** using **Athena** to deliver actionable insights through visualization and advanced metrics.

Aug'15 – Dec'16 | Pramati Technologies, India | Senior Software Engineer

Stack - Ruby on Rails, Memcached, Redis, Solr, Postgres

Key Result Areas:

- Developed a web application for managing patents with a robust role-based content management system for content manipulation, authorization, and access; additionally, organized and facilitated Open Source Meetups to foster community engagement and collaboration.

Jan'14 – Jul'15 | Tata Consultancy Services, India | Assistant System Engineer

Stack - Ruby on Rails, Memcached, Redis, Solr, Postgres

Key Result Areas:

- Headed a team of two members to develop a collaborative platform for sharing content, ideas, and processes across the Tata group of companies